PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GORVEMENT
FORM IV TAHOSSA NORTHWESTERN ZONE MOCK EXAMINATION 2025

032/1 **CHEMISTRY 1**

TIME 3:00HRS **AUGUST 2025**

INSTRUCTIONS

1. This paper consists of sections A and B and C
2. Answer all questions in sections A and B and two questions in section C.
3. Section A carries sixteen (16) marks, section B fifty four marks (54) and section C thirty marks (30).
4. Non-programmable calculator may be used.
5. Write your examination number on every page of your booklet(s).
6. The following constants may be used.
   Atomic masses H = 1, N = 14, Cu = 63.5, Cl = 35.5, Ca = 40,
   Ag = 108, Pb = 207, Fe = 56, Cu = 64
   Avogadro's number (LA) = 6.022×10^{23}
   G.M.V at S.T.P = $22.4dm^3$ or $22400cm^3$
   1 Faraday = 96500 coulombs
   1 litre = $1dm^3$ =$1000cm^3$

SECTION A: (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the most correct answer from the given alternatives and write its letter besides the item number in the answer booklet provided.

   (i) Any problem in the scientific research is affected by atleast two factors commonly referred to as variables. Which variables change its values when the value of the other variable change?

   A. Constant variable.
   B. Independent variable.
   C. Dependent variable
   D. manipulated variable
   E. Controlled variable.

   (ii) Which property is used to confirm the presence of pure water?

   A. It turns white anhydrous copper
   B. Sulphate blue
   C. It dissolves all other substances.
   D. It is neutral to litmus paper.
   E. It turns blue cobalt (II) chloride paper pink

   (iii)Bacteria can spoil food by breaking down organic matter. During this process, the releases unpleasant odors. Which of the following equations represent this spoilage?

   A. Bacteria + Food$\rightarrow$Methane+ Carbondioxide
   B. Mold + Sugar$\longrightarrow$Carbondixide+Alcohol.
   C. Food + Oxygen$\rightarrow$Water + carbodioxide

D. Organic matter + Bacteria ⟶ Simple organic + Odors molecules
E. Bacteria + sugar ⟶ Methane + Carbondioxide.

(iv) Why do we store fruits and vegetables in a refrigerator?
A. It improves their state.
B. The cold temperature kills bacteria
C. Cold temperature slow down the chemical reactions that cause spoilage.
D. It prevents them from drying out.
E. It makes to stay for a long time.

(v) What is the role of a Bunsen burner in a flame test experiment?
A. To create pleasant aroma in the lab
B. To identify metal ion by their characteristic flame colour.
C. To generate light during experiment.
D. To cool down the metal ion.
E. To display the types of flames.

(vi) Two solutions each50cm^3 with PH values of 1 and 9 respectively were mixed. What would be the PH value of the resulting solution?
A. Lower than 7
B. Equal to 10
C. Higher than 7
D. Equal to 100cm^3
E. Equal to 7

(vii) Regarding to the characteristics of a good fuel, sort out the correct statement.
A. It should have a lons pyrometric effect.
B. It should have high burning velocity.
C. It should have average ignition point.
D. It should have low burning velocity.
E. It should have low ignition point.

(viii) If some organic materials like timber, papers and fire wood catches fire suddenly, which kind of fire extinguisher will be best to stop fire?
A. Water.
B. Carbondioxide.
C. Fire blanket.
D. Dry powder.
E. Sand.

(ix) Which substance can be reduced when heated with carbon?
A. Aluminium oxide
B. Sodium oxide
C. Iron three oxide.
D. Calcium carbonate.
E. Magnesium oxide.

(x) Mr. Mwasu has got a tender to supply pure copper to a jewellary industry. He passed a current of 100000A through an electrolytic cell for half a day. What mass of pure copper was deposited at the cathode?
A. 1.421g.
B. 142.1g

C. 14.21kg
D. 142.13kg
E. 142000g.

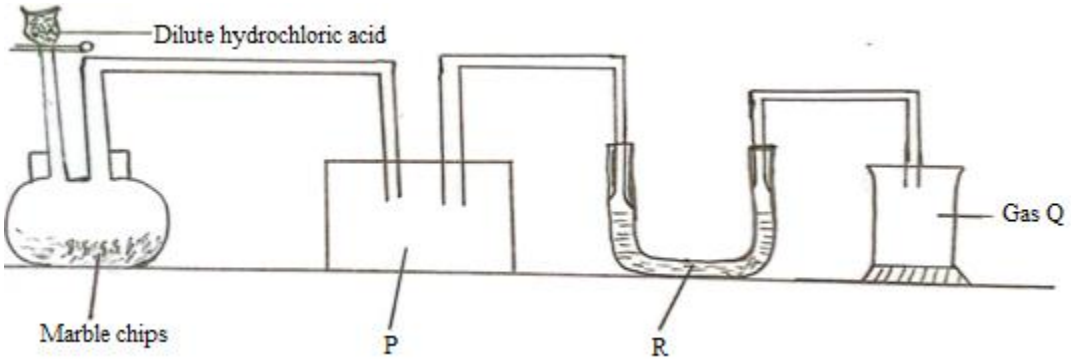
2. Match the items in List A with the correct response in List B by writing the letter of the correct response beside the corresponding item number in the answer booklet provided.

LIST A	LIST B
(i) Burns with a golden yellow flame	A. Lead
(ii) Bluish green flame	B. Calcium
(iii)Melts easily but show no definite colour flame	C. Zinc
(iv)Burns with yellow sparks flame	D. Iron
(v) Burns with a brick red flame	E. Potassium
(vi)Pale blue flame	F. Magnesium.
	G. Copper.
	H. Sodium.
	I. Silver.
	J. Methane.

SECTION B: (54 Marks)

Answer **all** questions in this section

3. The diagram below shows the laboratory preparation of gas Q, study the diagram and answer the questions that follows.



Questions

(a) (i) Name substance P and R
(ii) State the function of P and R.

(b) (i) Write the balanced chemical equation between marble chips and dilute hydrochloric acid.
(ii) Name the method used to collect gas Q in the diagram above and explain the reason for its suitability.

(c) By using chemical equation explain the chemical test of gas Q.

4. (a) It is not advised to use charcoal as a source of fuel due to its negative impacts to the environment. Support this statement by three points.
   (b) A form three student decided to boil water by using fire wood as a source of fuel. She had 6kg mass of charcoal fuel, 2 litres of water with 30˚C. After boiling, the charcoal and all the remained ashes were measured to be 2kg. Calculate the heat capacity of pure water ($C_w = 4.2KJkg^{-1}K^{-1}$, density of pure water is $1000kg^{-3}$)

5. (a) Briefly, given the meaning of the following terms as applied in chemistry
   (i) Molar mass
   (ii) Isotopes
   (iii)Standard solution
   (b) A chemistry teacher at Mawindi Secondary School guided his form three students to extract 20ml of orange juice from the orange fruit which contains Ethanoic acid. The students put 20ml of orange juice in the measuring cylinder of 100ml capacity where they added water up to 80ml mark of the cylinder 15.0ml of the dilute orange juice solution required 7.50ml of 0.2M of sodium hydroxide solution for complete neutralization reaction.

   (i) Write a balanced chemical equation for the reaction above
   (ii) Calculate the Molarity of Ethanoic acid
   - After dilution.
   - Before dilution.

(c) How many times does the orange juice be diluted?

6. (a) Explain the following terms as applied in electrolysis
   (i) Electrode
   (ii) Electrochemical equivalent
   (iii)Faraday's Constant.
   (b) Describe the mechanism electrolysis of copper (II) sulphate solution using copper electrodes (Diagram is not necessary)

   (c) 4.56 litres of oxygen gas measured at S.T.P were collected during electrolysis of copper (II) sulphate solution.

   Calculate the mass of copper deposited at the cathode.

7. (a) Chemical substance L turns a wet blue litmus paper red. It reacted with substance R which has bitter taste to produce compound W and compound M. Compound M turns white anhydrous copper (II) sulphate blue. Compound W can be obtained from the reaction mixture by evaporation. With six points, describe briefly the usefulness of compound W.
   (b) Basing on their P^H values, what are the three types of salt?

8. (a) Give the meaning of the following terms as applied in organic chemistry;
   (i) Functional group
   (ii) Locant

(iii)Addition reaction.

(b) Write all possible isomers of C_6H_{14} and give their names

(c)Show how Propane reacts with excess oxygen gas.

SECTION C: (30 Marks)

Answer **any two** questions from this section

9. Global warming is a huge problem resulting from environmental pollution. By using three points in each, explain its effects to the universe and remedial measures to combat the problem.

10. "A fertile soil is not necessarily productive" Justify his statement with the aid of six (6) points.

11. Extraction of metal is a reduction process. Explain briefly using two examples of metals above carbon and below carbon in the series about this fact.